



Two Sides Of The Same Coin

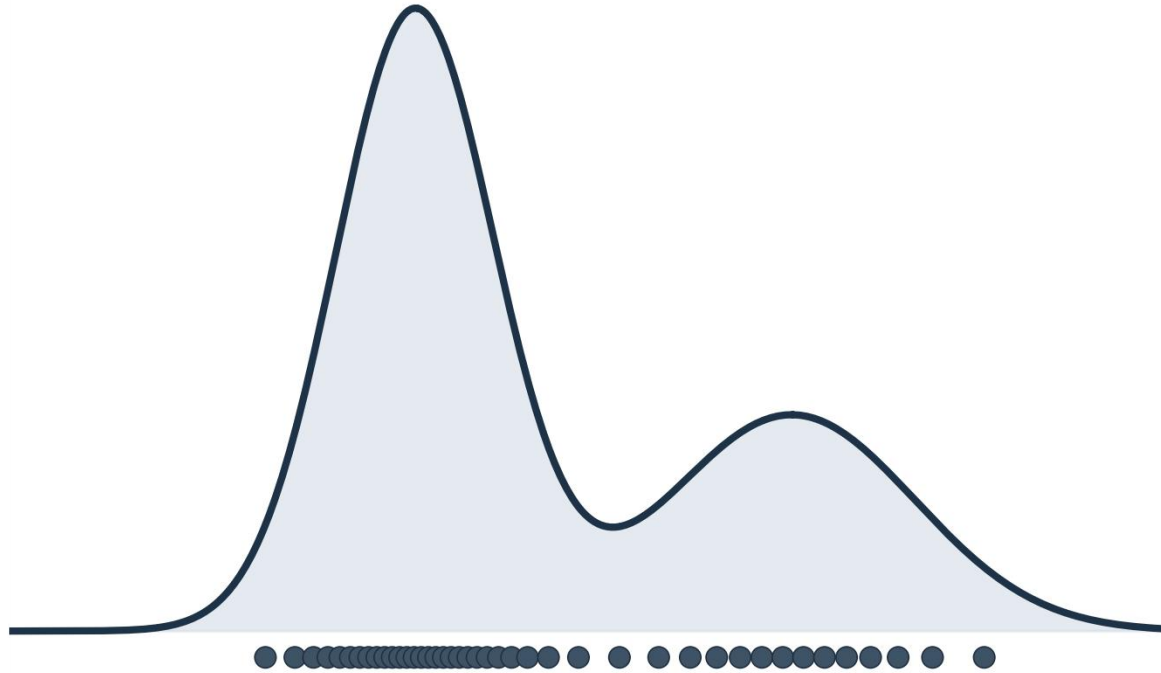
Weights & Positions



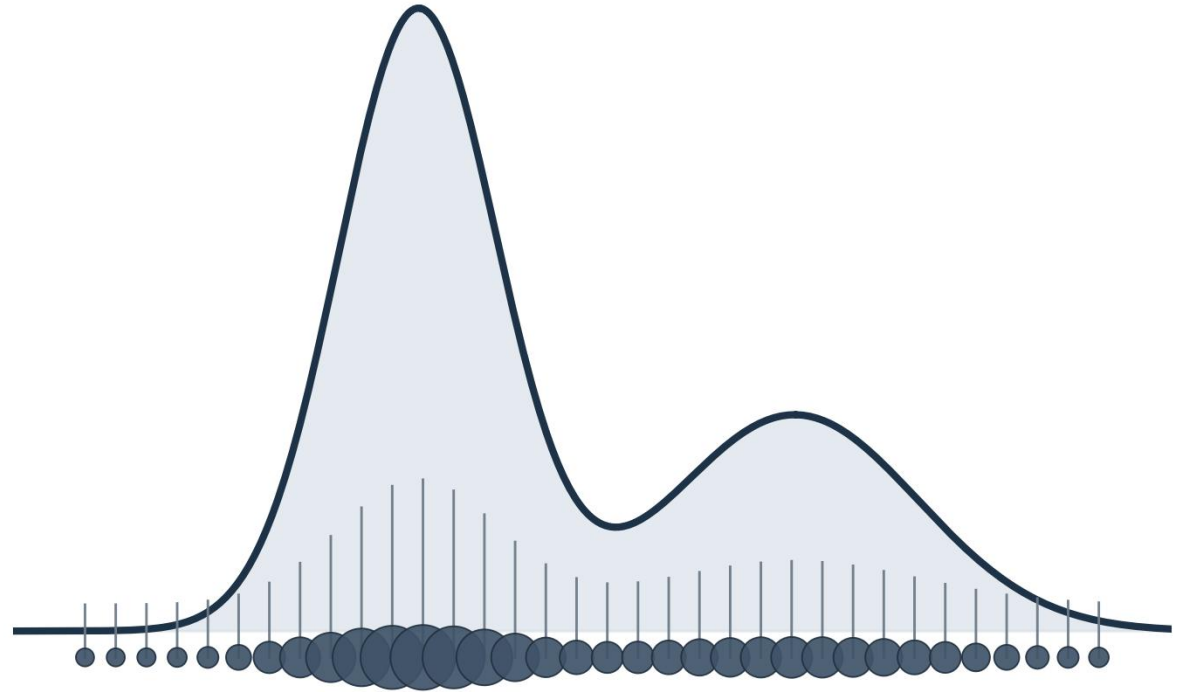
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$$\rho \approx \frac{1}{N} \sum_i \delta_{x^i}$$



$$\rho \approx \sum_j w_j \delta_{x^j}$$

Dynamics

Geometry

Optimality

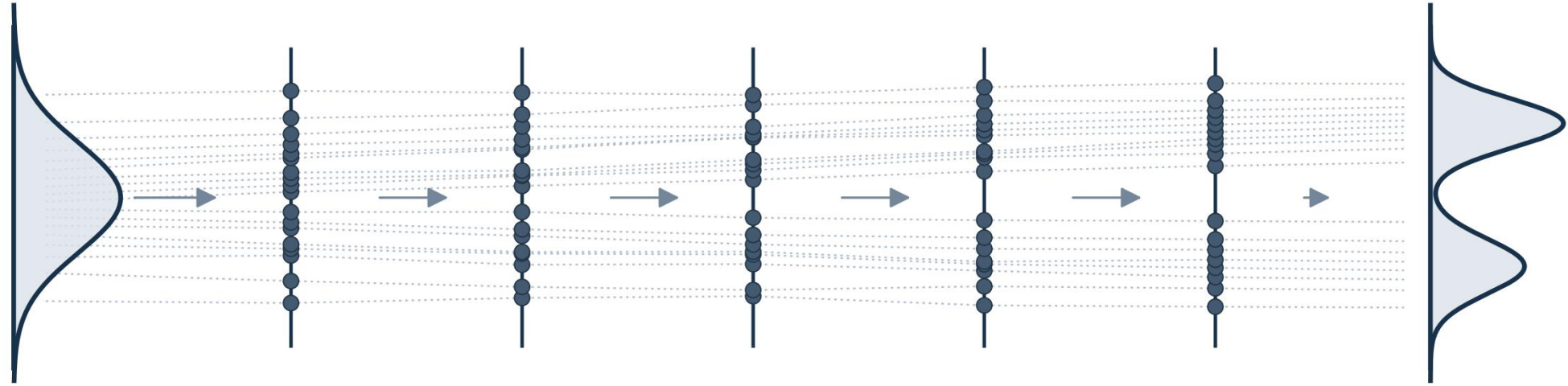
Connection with SVGD & MMD

Connection with PIC

Connection with Jarzynski equality

Weakness and Strengths

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Dynamics

Differential Equation

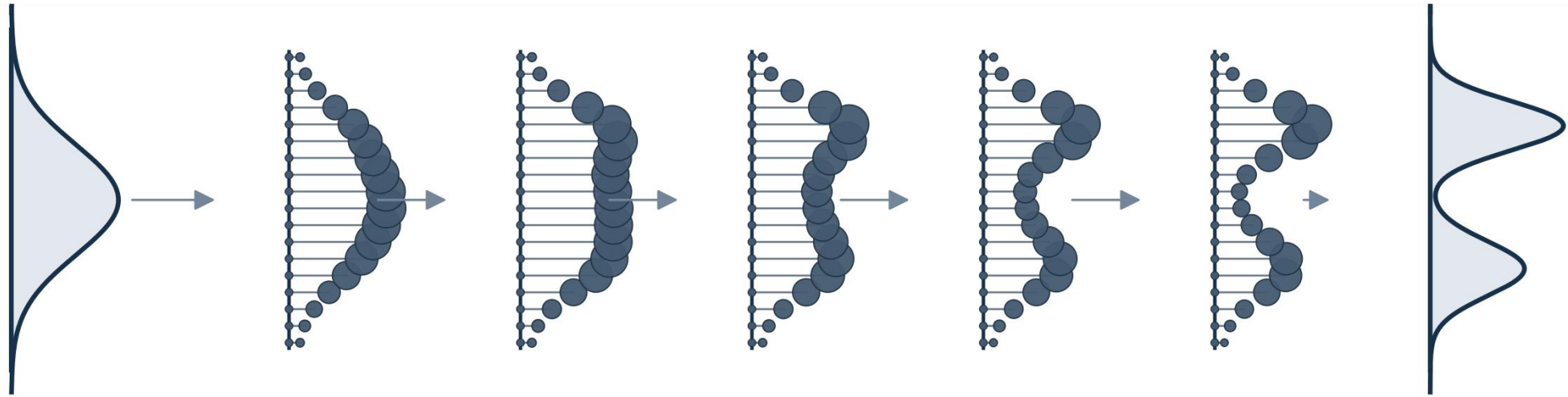
Deterministic

$$dX_t = v_t(X_t)dt$$

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0$$

Stochastic

$$dX_t = \sigma(X_t, t)dW_t, \quad \frac{\partial p(x, t)}{\partial t} = \frac{1}{2} \sum_{i,j} \frac{\partial^2}{\partial x_i \partial x_j} [(\sigma \sigma^\top)_{ij}(x, t)p(x, t)] .$$



Dynamics

Closed Form Weights

Differential Equation

Unbalanced $\frac{d}{dt} \log w_t = r_t$

$$w_t(X_t) = w_0(X_0) \exp \left(\int_0^t r_s(X_s) ds \right).$$

$$\partial_t \rho_t = \rho_t r_t$$

Balanced $\frac{d}{dt} \log w_t = r_t - \bar{r}_t$
 $\bar{r}_t = \int r_t(x) \rho_t(x) dx.$

$$w_t(X_t) = w_0(X_0) \exp \left(\int_0^t r_s(X_s) - \bar{r}_s ds \right).$$

$$\partial_t \rho_t = \rho_t (r_t - \bar{r}_t)$$

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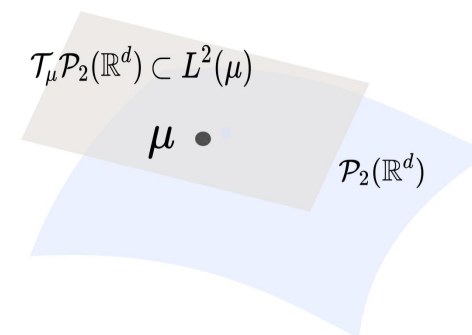
Content

Wasserstein
Geometry

$$W_2^2(\mu_0, \mu_1) = \inf_{\rho_t, v_t} \int_0^1 \int \|v_t(x)\|^2 \rho_t(x) dx dt.$$

$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = 0.$$

$$\rho_0 = \mu_0, \rho_1 = \mu_1$$

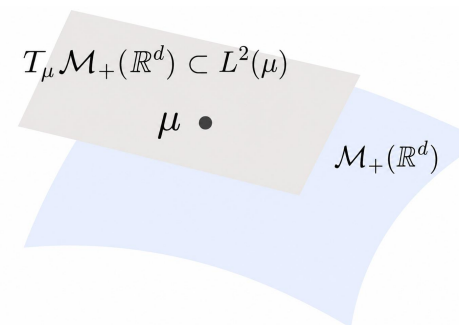


Fisher-Rao
Geometry

$$FR^2(\mu_0, \mu_1) = \inf_{\rho_t, r_t} \int_0^1 \int r_t(x)^2 \rho_t(x) dx dt.$$

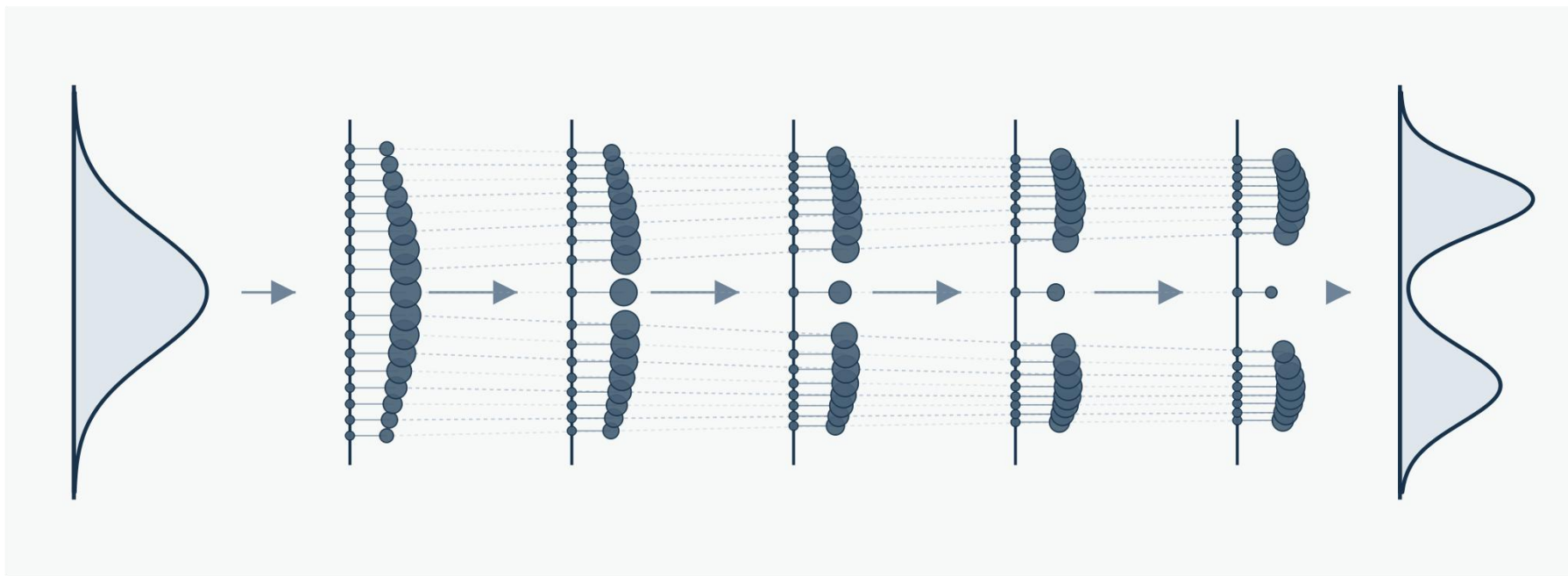
$$\partial_t \rho_t = \rho_t r_t.$$

$$\rho_0 = \mu_0, \rho_1 = \mu_1$$

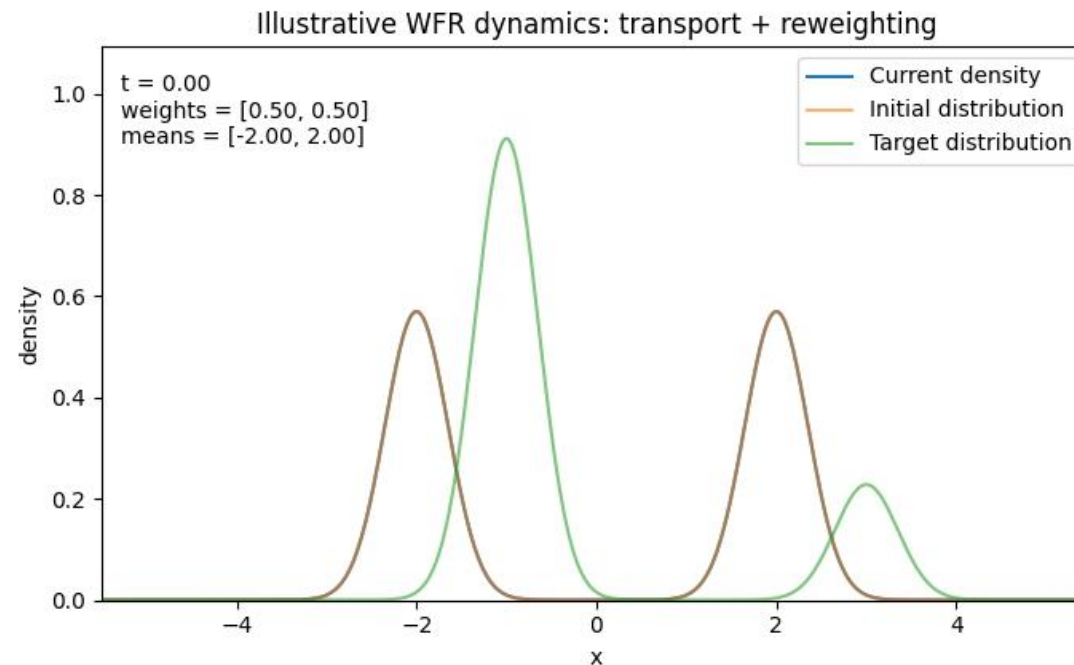
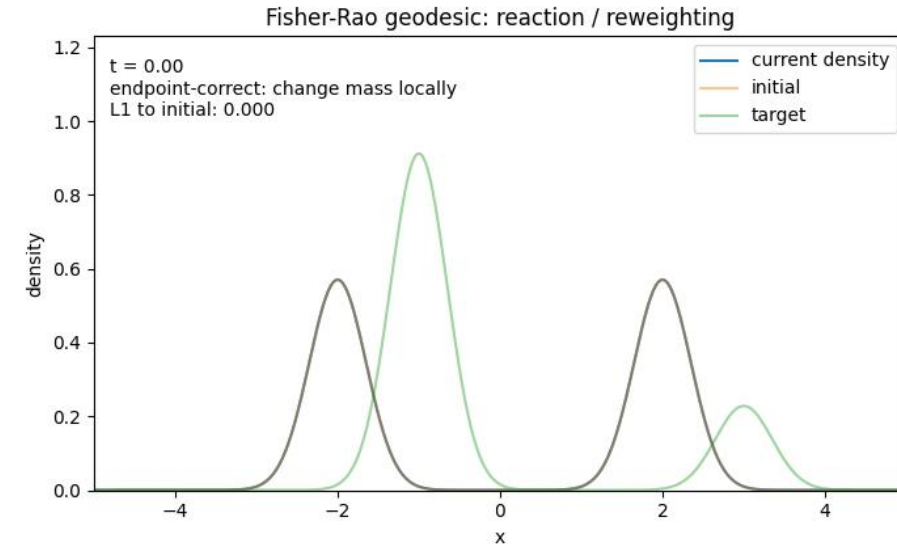
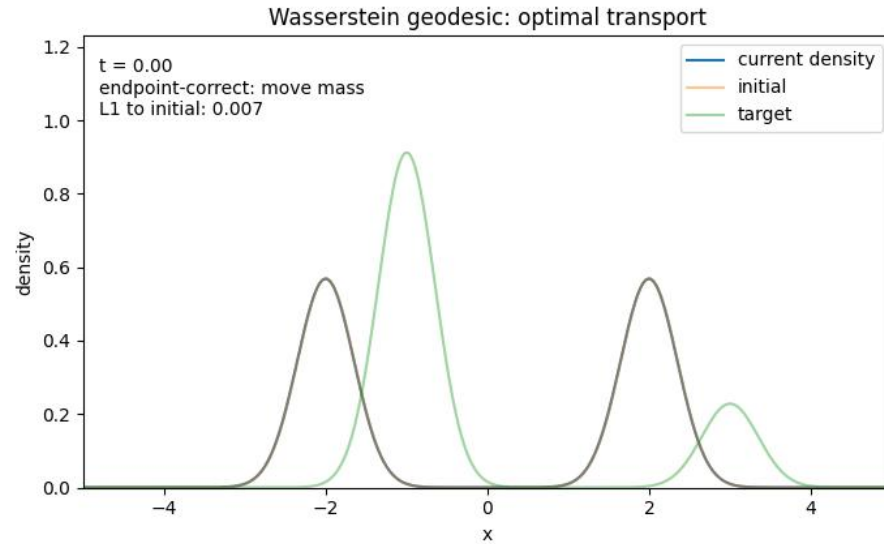


Wasserstein
Fisher-Rao
Geometry

$$WFR_2^2(\mu_0, \mu_1) = \inf_{\rho_t, v_t, r_t} \int_0^1 \int (\|v_t(x)\|^2 + \lambda^2 r_t(x)^2) \rho_t(x) dx dt.$$
$$\partial_t \rho_t + \nabla \cdot (\rho_t v_t) = \rho_t r_t.$$
$$\rho_0 = \mu_0, \rho_1 = \mu_1$$



Different Dynamics Comparison



Dynamics

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Variational Problem

Wasserstein $\rho^{k+1} \in \arg \min_{\rho \in \mathcal{P}_2(\Omega)} \left\{ \mathcal{F}(\rho) + \frac{1}{2\tau} W_2^2(\rho, \rho^k) \right\}.$

Fisher-Rao $\rho^{k+1} \in \arg \min_{\rho \in \mathcal{M}_+(\Omega)} \left\{ \mathcal{F}(\rho) + \frac{1}{2\tau} FR^2(\rho, \rho^k) \right\}.$

WFR $\rho^{k+1} \in \arg \min_{\rho \in \mathcal{M}_+(\Omega)} \left\{ \mathcal{F}(\rho) + \frac{1}{2\tau} WFR^2(\rho, \rho^k) \right\}.$

Gradient Flow

$$\partial_t \rho_t = \nabla \cdot \left(\rho_t \nabla \frac{\delta \mathcal{F}}{\delta \rho} [\rho_t] \right).$$

$$\partial_t \rho_t = -\rho_t \frac{\delta \mathcal{F}}{\delta \rho} [\rho_t].$$

$$\partial_t \rho_t = \nabla \cdot \left(\rho_t \nabla \frac{\delta \mathcal{F}}{\delta \rho} [\rho_t] \right) - \rho_t \frac{\delta \mathcal{F}}{\delta \rho} [\rho_t].$$

$\mathcal{F}(\rho)$	$\phi = \frac{\delta \mathcal{F}}{\delta \rho}$	Wasserstein gradient flow part WFR gradient flow
$\int V \rho \, dx$	V	$\partial_t \rho = \nabla \cdot (\rho \nabla V) - \rho V$
$\int \rho \log \rho \, dx$	$\log \rho + 1$	$\partial_t \rho = \Delta \rho - \rho(\log \rho + 1)$
$\text{KL}(\rho \parallel \pi)$	$\log \frac{\rho}{\pi} + 1$	$\partial_t \rho = \nabla \cdot \left(\rho \nabla \log \frac{\rho}{\pi} \right) - \rho \left(\log \frac{\rho}{\pi} + 1 \right)$
$\int \rho \log \rho \, dx + \int U \rho \, dx$	$\log \rho + U + 1$	$\partial_t \rho = \Delta \rho + \nabla \cdot (\rho \nabla U) - \rho(\log \rho + U + 1)$
$\frac{1}{2} \int (\rho - \pi)^2 \, dx$	$\rho - \pi$	$\partial_t \rho = \nabla \cdot (\rho \nabla (\rho - \pi)) - \rho(\rho - \pi)$
$\frac{1}{2} \text{MMD}_k^2(\rho, \pi)$	h_ρ	$\partial_t \rho = \nabla \cdot (\rho \nabla h_\rho) - \rho h_\rho$

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An aerial photograph of a modern campus, likely a university, featuring a large stadium with a curved roof, several multi-story buildings, and extensive green spaces with trees. The image is overlaid with a semi-transparent blue filter.

Content

MMD $\xrightarrow{\text{kernelize force}}$ Fisher-Rao

$$\partial_t \mu = -K^{-1} \mu \frac{\delta \mathcal{F}}{\delta \mu}$$

$$\partial_t \mu = -\mu \frac{\delta \mathcal{F}}{\delta \mu}$$

Wasserstein $\xrightarrow{\text{kernelize velocity}}$ Stein.

$$\partial_t \mu = \nabla \cdot \left(\mu \nabla \frac{\delta \mathcal{F}}{\delta \mu} \right).$$

$$\partial_t \mu = \nabla \cdot \left(\mu K_\mu \nabla \frac{\delta \mathcal{F}}{\delta \mu} \right)$$

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An aerial photograph of a modern campus, likely a university, featuring several buildings with green roofs, a large sports field, and a river winding through the landscape. The image is overlaid with a semi-transparent blue filter.

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Optimization Formulation

$$\begin{aligned} \min_u \mathbb{E} & \left[\int_0^1 \left(\frac{1}{2} \|u(X_t^u, t)\|^2 + f(X_t^u, t) \right) dt + g(X_1^u) \right], & \text{Objective Function} \\ \text{s.t. } & \underline{dX_t^u = (b(X_t^u, t) + u(X_t^u, t))dt + \sigma(t)dB_t}, & \text{State Transition Equation} \\ & \underline{X_0^u \sim p_0} & \text{Initial Probability Distribution (fixed)} \end{aligned}$$

Reweighting Formulation

$$W(X_{t:T}^0) = \exp \left(-\frac{1}{\lambda} \left[\int_t^T f(X_s^0, s) ds + g(X_T^0) \right] \right) \quad dX_t^0 = b(X_t^0, t) dt + \sigma(t) dB_t$$
$$\mathbb{P}^*(X_{t:T}^0) = \frac{W(X_{t:T}^0) \mathbb{P}_0(X_{t:T}^0)}{\mathbb{E}_{\mathbb{P}_0}[X_{t:T}^0]}$$

Dynamical Reweighting Formulation

$$\mu_t(x) = \mathbb{E}_{\mathbb{P}_0}[\bar{w}_t \delta(X_t = x)],$$

$$\partial_t \mu_t = -\nabla \cdot (b \mu_t) + \frac{1}{2} \nabla \nabla : (\sigma \sigma^\top \mu_t) - \frac{1}{\lambda} f(x, t) \mu_t,$$

Path Integral Control

$$u^*(x, t) = \nabla_x \log \mathbb{E}_{X \sim p^{base}} \left[\exp \left(- \int_t^T f(X_s, s) ds - g(X_T) \right) \middle| X_t = x \right]$$
$$dX_t^u = (b(X_t^u, t) + \sigma(X_t^u, t) u^*(x, t)) dt + \sigma(X_t^u, t) dW_t,$$

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Proposition 1 (Jarzynski equality). *Let (X_t, A_t) solve the coupled system of SDE/ODE*

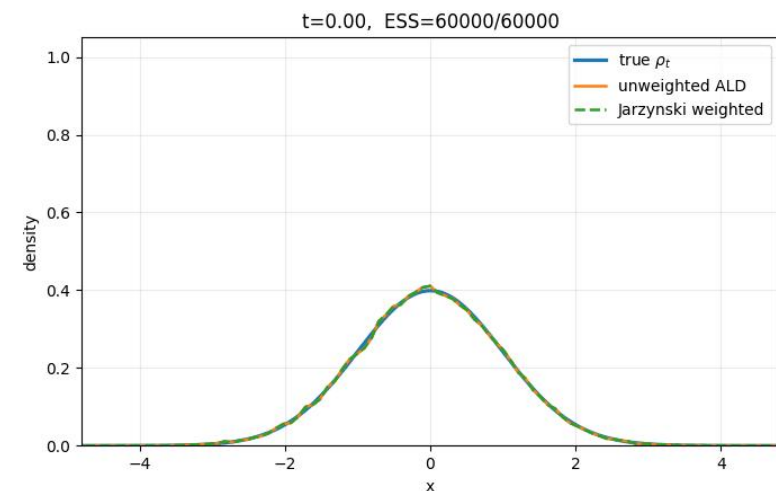
$$dX_t = -\varepsilon_t \nabla U_t(X_t) dt + \sqrt{2\varepsilon_t} dW_t, \quad X_0 \sim \rho_0, \quad (10)$$

$$dA_t = -\partial_t U_t(X_t) dt, \quad A_0 = 0, \quad (11)$$

where $\varepsilon_t \geq 0$ is a time-dependent diffusion coefficient and $W_t \in \mathbb{R}^d$ is the Wiener process. Then for all $t \in [0, 1]$ and any test function $h : \mathbb{R}^d \rightarrow \mathbb{R}$, we have

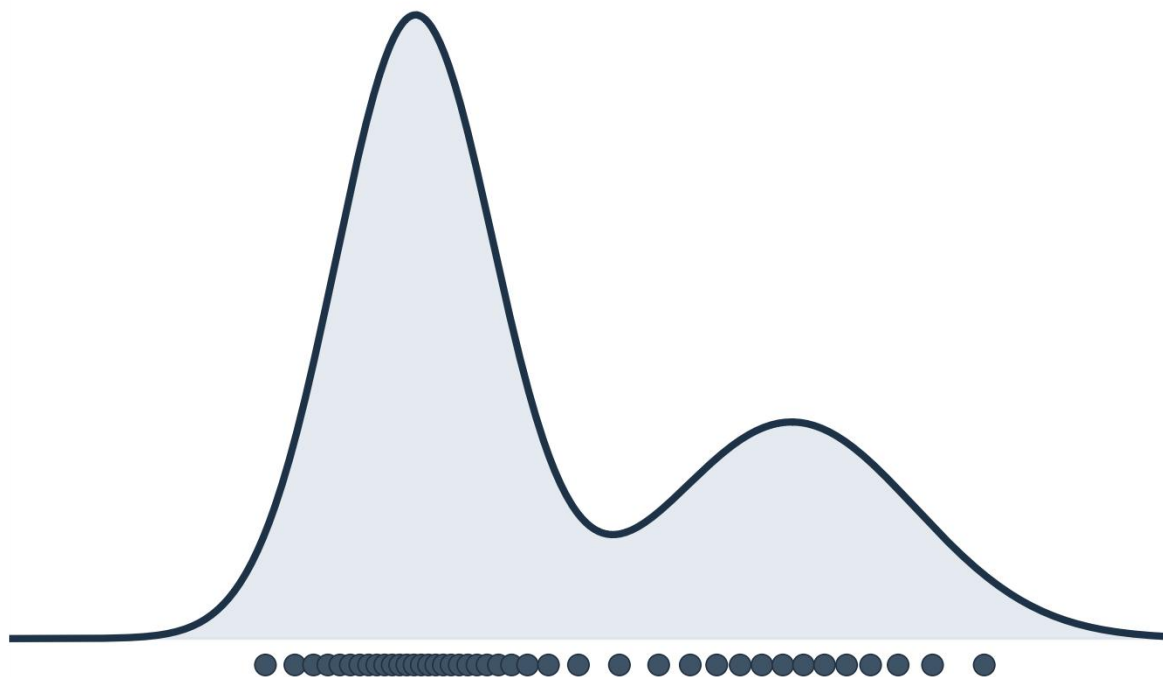
$$\int_{\mathbb{R}^d} h(x) \rho_t(x) dx = \frac{\mathbb{E}[e^{A_t} h(X_t)]}{\mathbb{E}[e^{A_t}]}, \quad Z_t/Z_0 = e^{-F_t + F_0} = \mathbb{E}[e^{A_t}], \quad (12)$$

where the expectations are taken over the law of (X_t, A_t) .

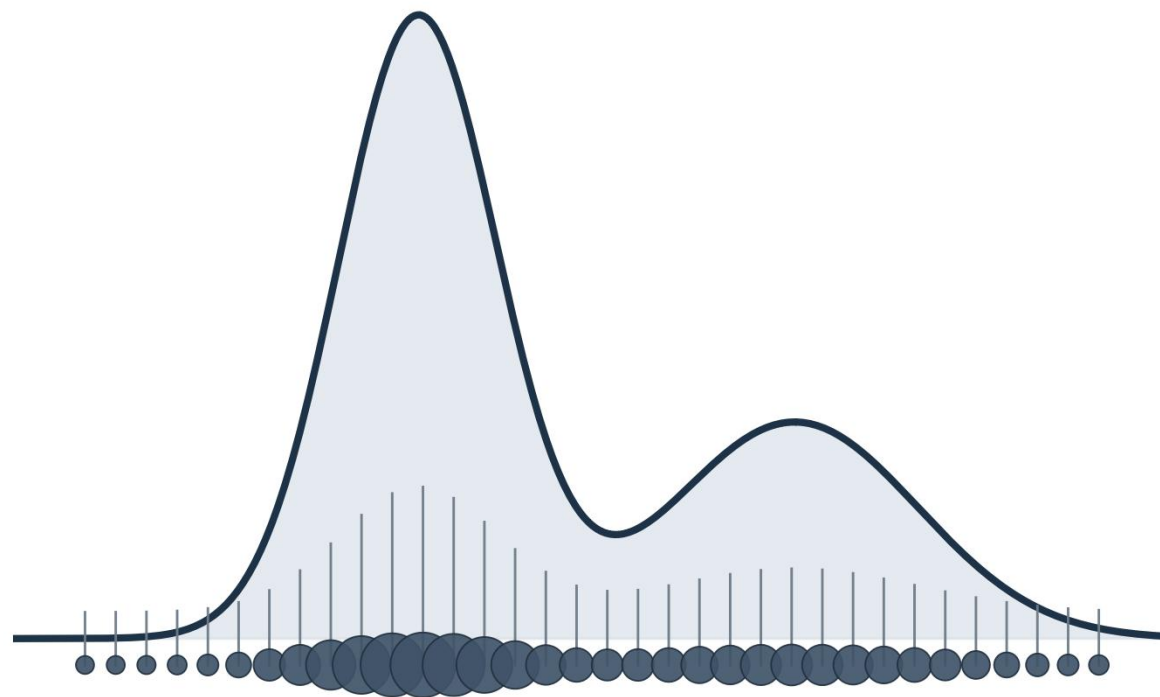


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适合有采样点的情况
理论最优，不会出现退化
计算复杂，通常需要优化



适合有密度函数的情况
会出现退化情况
计算简单，可以快速修正已有的采样点